

Abelian

$ab = ba \rightarrow \text{group} \rightarrow \Sigma$

$\frac{\langle a, b | ab=ba \rangle}{\mathbb{Z} \times \mathbb{Z}}$

$S^1 \vee S^2$

$T = \mathbb{Z} \times \mathbb{Z}$

$K(\cdot, \mathbb{Z} * \mathbb{Z})$

[illegible]

R/P^2 :
0-cell: $\{v\}$
1-cell: $\{\gamma\}$
2-cell: $\{2\gamma\}$

R/P^2 : $\mathbb{Z}_2$ $\langle a | a^2 \rangle$
 $a^2 = 1$

R/P :
0-cell: $\{v\}$
1-cell: $\{\gamma\}$
2-cell: $\{2\gamma\}$

$\pi_1(R/P) = \pi_1(\mathbb{RP}^1) = \mathbb{Z}_2$

Hand-drawn sketches on graph paper:

- A sequence of five circles of increasing size, arranged in a horizontal row.
- A flower-like shape with five petals, enclosed in a circle.
- A spiral shape, resembling a nautilus shell, enclosed in a circle.
- A horizontal line with four arrows pointing to the right.
- A complex plane with a branch cut along the real axis, showing a multi-valued function.
- A surface plot, possibly representing a function of two variables.
- A contour integral in the complex plane, with a branch cut and a contour around it.